

Spring 2026 Project

Electronics Tool Bench

Description

Technology advances have allowed miniaturization and cost reductions in many areas. For example, our labs in Driftmier 1450 and 1409 have testbench equipment like oscilloscopes, function generators, power supplies and multimeters. The capabilities of this equipment were once relegated to stand alone, custom built devices. Today, you can find many of these capabilities in low-cost, USB, microcontroller based devices. At the heart of these testbench devices lie the basic (and advanced, of course) electronic circuits used in many designs. In addition, the interaction of these devices with the person using them requires a concise user interface (UI) to control the device and interpret data.

The primary objective of your team's project is to design a set of electronic testbed devices using your Raspberry PI as the single-board computer controlling the function, user interface, and data acquisition features specified in the design. This is called the ECSE 2920 Electronics Tool Bench.

Your team will be provided a kit of parts to use. Our objective is to keep your cost to a minimum. However, any additional components that your team may need that are not supplied to you is your team's responsibility, and if your team "cooks" a key component (see list), **your team must replace the component or receive an incomplete for the course.**

IMPORTANT: This document is a big picture view of the project. Your team will especially reference it throughout the semester when you are making larger design decisions. To aid in helping your team to complete the objectives, weekly checkpoints will be provided via the ECSE 2920 repository on Github (which you will need to refer to often). These checkpoints will need to be turned in weekly to the corresponding ELC folder.

Disclaimer: This document may be updated throughout the semester as this is a dynamic project. If it is, it will be announced to the class in a timely manner.

Design Constraints/Boundary Conditions

- 1) Your team may only use components given to you for "active" circuitry in your design (transistors, op-amps, processor, convertors, etc...). If you would like to use an active component not provided, your team will need to get permission and your team will be responsible for additional costs.

- 2) A 24V AC/DC supply is provided. Your team must derive/design all additional voltage values from this supply *only*. *Note*: The raspberry Pi is powered by the USB-C connector provided. (However, as you'll learn, the Pi Protection Hat (PPH) will not supply large currents.)
- 3) As described in the syllabus, your team's project will need to achieve various grade-level checkpoints. These checkpoints are described in this document.
- 4) Your team will write code using Python and the Raspberry PI4 provided to you. No additional microcontrollers or microprocessors are allowed.
- 5) For AI constraints refer to the syllabus and comments made in-class.

Grade Level Descriptions

Note: Each of the key objectives are represented by weekly deliverables on the course schedule (on Github). Each will be graded based on a rubric which will be provided. Please refer to the syllabus and additional information from the instructors for group and individual semester grades.

Minimum for a "C"

Your team's electronics tool bench is a semester-long journey which starts with a few steps:

- 1) Verify your team's Pi-Protection-Hat (PPH). This must be used during the entire project.
- 2) Characterize the digital potentiometers used on this project:
 - a. Give a concrete description of what they do and how they work and initialize.
 - b. Include documentation showing that you programmed the potentiometer to multiple values (100, 1000, 5000, 10000) Ohms.
- 3) The team's user interface (UI) is working. Use the Raspberry PI, LCD, and a single dual digital potentiometer to program a variable resistance.
 - a. The digital encoder (knob) will control the resistance value desired from the digital potentiometer.
 - b. Your team must have a "slow" and a "fast" detector for the knob spin. For slow, increment the value shown to the user by "10 Ω ". For fast, increment the value shown to the user by "100 Ω ".
 - i. Your team must use the final value to program the digital potentiometer to change to this value as a resistance.
 - ii. The user should use the push button feature on the knob to set the value and program the potentiometer.
 - iii. The current value should be shown on the LCD. In testing this will be verified with a multimeter.
 - c. The output resistance of the potentiometer should match your selected value.
 - d. Your team will also need to demonstrate both potentiometers on the IC chip working individually. Use your LCD display and knob input to choose which potentiometer to program. You must be able to program each independently without changing the other. For example, you should have a primary menu to select which potentiometer to program (a knob spin highlights the desired potentiometer, and pressing the knob will select the

desired potentiometer). After choosing a potentiometer, you should be taken to a value entry menu; here, you will spin the knob and browse through the desired resistance values (with the criteria described above). Finally, implement code to take you back to the primary menu by holding the knob for three (3) seconds or longer.

- e. In regards to the DFRobot LCD screen, the buttons are not to be used. Selection within the UI should be done using the rotary encoder only.

This document is intended for guiding principles, for additional specifications, please see the Checkpoint C rubric. Note this must be passed with a C or better.

Minimum for a “B”

All of the above plus the following functionality... For the following all voltage levels are assumed to be peak to peak.

1) Function Generator:

- a. Produce an X-axis centered *square wave* with adjustability:
 - i. Use the UI to produce a variable amplitude within the range of +/- 10V.
 - 1. The square wave should be centered at 0V.
 - 2. The resolution for x-axis and peak values should be within 0.3125V and all values should shift in the same relative direction.
 - ii. Use the UI to change the frequency to any value from 100Hz-10KHz.
 - 1. The duty cycle should be 50%.
- b. Use UI to display the live values being outputted.
- c. The function generator should not drive current and should be considered High Z.

2) Ohmmeter:

- a. Measures from 500-10K Ω range from external resistance input and display to the user how close to the true value they are (i.e. 50 Ω +/- Y Ω). Y should be representative of the value that is true to your design. This should be derived in your documentation. The Ohmmeter should be autoranging.
- b. Use UI to report back values to user.

3) Voltmeter:

- a. Must be able to measure a DC voltage from +/- 5V and display to the user how close to the true value they are (i.e. 2.5V +/- **0.3125V**).
- b. Use UI to report back values to user.
- c. Must be able to measure voltage from an external supply OR from your DC voltage reference (see next).

4) DC Voltage Reference

- a. Measure with High Z.
- b. This is a reference which does not supply high current.
- c. Output is +/- 5V in 0.625V steps.
- d. Use UI to control the output.
- e. Use UI to report back values to user.

- 5) Completed UI (NOTE: Your UI *MUST* follow this tree, deviations are not allowed. Color creations are allowed 😊)
- a. UI must use the rotating feature to scroll through selections and the push button feature to select/input. Each selectable area must have a return to Main and a Back select. The back select takes the UI back one in the tree.
 - b. The Main level UI is:
 - i. OFF: Everything is off.
 - ii. Mode Select
 1. Function Generator
 - a. Type
 - i. Square
 - ii. Back
 - iii. Main
 - b. Frequency
 - i. Input Frequency
 - ii. Back
 - iii. Main
 - c. Amplitude
 - i. Input Amplitude
 - ii. Back
 - iii. Main
 - d. Output
 - i. On (Note: if you back out of this or go to main, must turn off by default). UI must show value of generated output.
 - ii. Off
 - iii. Back
 - iv. Main
 2. Ohmmeter (UI shows value of reading and threshold)
 - a. Back
 - b. Main
 3. Voltmeter (UI show value of reading and show threshold)
 - a. Source
 - i. External
 - ii. Internal Reference (If selected go to DC Reference)
 - iii. Back
 - iv. Main
 4. DC Reference
 - a. Voltage Value Input
 - b. Output
 - i. On (note if you back out of this or go to main, must turn off by default). UI must show value of generated output. UI must show value of generated output using your Voltmeter!)

- ii. Off
 - iii. Back
 - iv. Main
 - c. Back
 - d. Main
- 5. Back
- 6. Main

This document is intended for guiding principles, for additional specifications, please see the Checkpoint B rubric. Note this must be passed with a C or better. Additional specifications will be provided about PCB completion.

Minimum for an "A-"

All of the above plus the following functionality...

Your team will now add a Sine wave function generator and a simple frequency measuring feature. This is not an oscilloscope because you are not drawing the waveform. You only need to measure the frequency.

- 1) Sine Wave Generator
 - a. Positive voltage only, 0-10V, variable amplitude in 0.625V steps.
 - b. Frequency is variable from 1K-10KHz, variable by 500Hz.
 - c. Same High Z assumptions apply.
- 2) Sine Wave Frequency Measurement
 - a. 1K-10KHz input range. Accuracy affects score.
 - b. Measure an input sine wave from a different device.
- 3) This checkpoint changes the UI (**changes in bold**):
 - a. The Main level UI is:
 - i. OFF: Everything is off
 - ii. Mode Select
 1. Function Generator
 - a. Type
 - i. **Sine**
 - ii. Square
 - iii. Back
 - iv. Main
 - b. Frequency
 - i. Input Frequency
 - ii. Back
 - iii. Main
 - c. Amplitude
 - i. Input Amplitude
 - ii. Back
 - iii. Main
 - d. Output

- i. On (note if you back out of this or go to main, must turn off by default). UI must show value of generated output.
 - ii. Off
 - iii. Back
 - iv. Main
 - 2. Ohmmeter (UI shows value of reading)
 - a. Back
 - b. Main
 - 3. Voltmeter (UI show value of reading)
 - a. Source
 - i. External
 - ii. Internal Reference (If selected, go to DC Reference)
 - iii. Back
 - iv. Main
 - 4. DC Reference
 - a. Voltage Value Input
 - b. Output
 - i. On (note if you back out of this or go to main, must turn off by default). UI must show value of generated output using your Voltmeter!)
 - ii. Off
 - iii. Back
 - iv. Main
 - c. Back
 - d. Main
 - 5. Frequency Measurement**
 - a. Back**
 - b. Main**
 - 6. Back
 - 7. Main

This document is intended for guiding principles, for additional specifications, please see the Checkpoint A- rubric. Note this must be passed with a C or better.